Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (CS) SEM-I EXAMINATION (CBCS)

U61A5FM: FUNDAMENTAL OF MATHEMATICS

JAN - FEB 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

Q-1. **Answer the following:** (Any Five) **(30)**

A. Explain UNION, INTERSCT and MINUS SET operation in brief.

B. Explain different methods for SET representation with example.

C. Explain Row matrix, Column matrix, Square matrix, Equal matrix, null matrix with example.

D. Discuss the Homeomorphic Graph in details

E. Explain various types of functions with example.

F. How many no of combination has for Group of four can be done from 12 people?

G. Create Pascal's triangle up to $(a+b)^5$

SECTION-II

Q-2. **Answer the following: (Any Two)** **(12)**

I. Prove De' morgan law:

1. **$(A \cup B)' = A' \cap B'$**

$(A \cap B)' = A' \cup B'$

II. $A=\begin{bmatrix}5 & 1\\3 & 4\end{bmatrix}$ $B=\begin{bmatrix}2 & 1\\5 & 2\end{bmatrix}$ calculate $A \times B$, $B \times A$

III. A={1,2,3,4,5,6,7}, B={3,4,7,8}, C={1,7,9} find out $A \cup B$, $A \cup (B \cap C)$, $(A \cap C) \cup B$

Q-3. **Answer the following: (Any Two)** **(12)**

I. Define following with example: **Directed graph, Pendent vertex, Cycle path.**

II. Explain domain, co domain, Range in function with example.

III. Discuss Isomorphic Graph in detail.

Q-4. **Answer the following: (Any Two)** **(6)**

I. Find the number of combinations of five object a,b,c,d,e taken three at a time.

II. Find the number of words, with or without meaning that can be formed with the letters of the word 'BUBBLE'.

III. Explain pigeonhole principle in detail.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. IT (Cyber Security) SEM-I EXAMINATION

U61B6CS1: LANGUAGE AND COMMUNICATION SKILLS – I

Time: 3 Hours] JAN - FEB 2023 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. ATTEMPT ANY FIVE OUT OF SEVEN. **(30)**

1. Explain the functions of communication with an example.
2. Discuss the extrapersonal and intrapersonal communication in your own words with an example.
3. Illustrate the techniques of reading with an example.
4. Classify the traits of Good Listener.
5. List the barriers of listening with an example in your own words.
6. Summarize the difference between listening and hearing with an example.
7. Define communication and explain the Process of communication with diagram.

SECTION-II

2. (A) Answer the following: (Any One) **(6)**

1. Discuss the non verbal communication in details with its sub part.
2. What do you understand by the term inter cultural communication in organization? Explain it with sub parts.

(B) Fill in the blanks using appropriate answer from the options given in the brackets. (All are compulsory) **(6)**

1 The word Communication is derived from the _____.

(A) America (B) England (C) Latin

2 Kinesics is a part of _____ communication.

(A) Verbal (B) Non-verbal (C) both A & B

3 The messages that you send with your voice are known as _____.

(A) Paralanguage (B) Nonverbal (C) None of these

4 He didn't work hard _____ he failed his exam.

(A) So (B) And (C) Because

PTO